

Euclidean Inner-product

$$x = (x_1, \dots, x_n); y = (y_1, \dots, y_n) \in \mathbb{R}^n$$

$$x \cdot y = x_1 y_1 + \dots + x_n y_n$$

→ Properties: $x, y, z \in \mathbb{R}^n, u \in \mathbb{R}$

$$1) x \cdot y = y \cdot x$$

$$2) (x+y) \cdot z = z \cdot x + z \cdot y$$

$$3) u(x \cdot y) = (ux) \cdot y$$

$$4) x \cdot x \geq 0$$

$$x \cdot x = 0 \text{ iff } x = 0$$

Euclidean norm

$$x = (x_1, \dots, x_n) \in \mathbb{R}^n$$

$$\|x\| = \sqrt{x \cdot x} = \sqrt{x_1^2 + \dots + x_n^2}$$

→ Properties:

$$1) \|x\| \geq 0, \|x\| = 0 \text{ iff } x = 0$$

$$2) \|ux\| = |u| \cdot \|x\|$$

$$3) \|x+y\| \leq \|x\| + \|y\|$$

~ To be a norm it has to verify all these properties

Inner-Product

$$x \cdot y = \|x\| \cdot \|y\| \cdot \cos \theta \quad (\Rightarrow)$$

$$(\Rightarrow) \frac{x \cdot y}{\|x\| \cdot \|y\|} = \cos \theta$$

$$\widehat{x \cdot y} = \frac{x \cdot y}{\|x\| \cdot \|y\|}$$

$$\|(x, y)\| (\Rightarrow) \sqrt{(x^2 + y^2)^2} = 1 (\Rightarrow) x^2 + y^2 = 1$$

$$\|(x, y)\|_M = 1 (\Rightarrow) |x| = 1 \wedge |y| \leq 1 \vee |x| \leq 1 \wedge |y| = 1$$

$$\begin{aligned} \|(x, y)\|_S = 1 (\Rightarrow) & (y = 1 - x \wedge x, y \geq 0) \vee \\ & (y = 1 + x \wedge y \geq 0 \wedge x \leq 0) \vee \\ & (y = -1 + x \wedge y \leq 0 \wedge x \geq 0) \vee \\ & (y = -1 - x \wedge x, y \leq 0) \end{aligned}$$

Cauchy Schwarz inequality

$$|x \cdot y| = \|x\| \cdot \|y\| \cdot \cos \theta$$

$$\frac{|x \cdot y|}{|\cos \theta|} = \|x\| \cdot \|y\|$$

$$\text{Then } |x \cdot y| \leq \|x\| \cdot \|y\|$$

as $|\cos \theta|$ between 0 and 1

Others norms

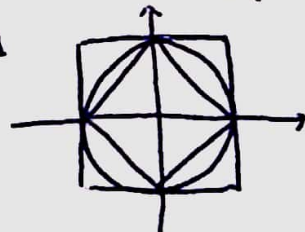
$$\|x\|_M = \max\{|x_1|, \dots, |x_n|\}$$

↳ norm of the maximum (chooses the maximum value among the modulus of the different values)

$$\|x\|_S = |x_1| + \dots + |x_n|$$

↳ norm of the sum

Adds up all the absolute values of x



$$\bigcirc \rightarrow \|x\| = 1$$

$$\square \rightarrow \|x\|_M = 1$$

$$\diamond \rightarrow \|x\|_S = 1$$

How to prove if something is a norm

- $\|(x, y)\| \geq 0$
- $\|u(x, y)\| = |u| \|(x, y)\|$
- $\|(x_1, y_1) + (x_2, y_2)\| \leq \|(x_1, y_1)\| + \|(x_2, y_2)\|$

Equivalence of a norm

$\|\cdot\|_1$ is equivalent to $\|\cdot\|_2$ if and only if there exist positive constants, m and M , such that

$$m\|\cdot\|_2 \leq \|\cdot\|_1 \leq M\|\cdot\|_2 \quad (1)$$

$$(1) \Rightarrow \frac{1}{M}\|\cdot\|_1 \leq \|\cdot\|_2 \leq \frac{1}{m}\|\cdot\|_1$$

For example:

$$\begin{aligned} \|(x, y)\|_1 &= |x| + |y| \leq 2|x| + 2|y| = \\ &= 2(|x| + |y|) = 2\|(x, y)\|_2 \end{aligned}$$

$$\begin{aligned} \|(x, y)\|_1 &= |x| + |y| \geq \frac{1}{2}|x| + \frac{1}{2}|y| = \\ &= \frac{1}{2}(|x| + |y|) = \frac{1}{2}\|(x, y)\|_2 \end{aligned}$$

$$\frac{1}{2}\|(x, y)\|_2 \leq \|(x, y)\|_1 \leq 2\|(x, y)\|_2$$

Level sets

$$\begin{aligned} f: D \subset \mathbb{R}^2 &\rightarrow \mathbb{R} \\ (x, y) &\rightarrow f(x, y) \end{aligned}$$

$$\text{Graph}(f) = \{(x, y, t) \in \mathbb{R}^3 : (x, y) \in D, t = f(x, y)\}$$

$$\text{Graph}(f) \subset \mathbb{R}^3$$

$$L_u = \{(x, y) \in D_f : f(x, y) = u\}, \text{ where } u \text{ is the level set of } f$$

The union of level sets is defined to be a contour map of f with interval p

$$\bigcup_{u \in p} f^{-1}(\lambda u p) = \bigcup_{u \in p} L_u$$

Euclidean Distance

$$x = (x_1, \dots, x_n)$$

$$y = (y_1, \dots, y_n)$$

$$\begin{aligned} d(x, y) &= \sqrt{(x_1 - y_1)^2 + \dots + (x_n - y_n)^2} \\ &= \|x - y\| \end{aligned}$$

The distance verifies the following conditions:

a) positiveness:

- $d(x, y) \geq 0, \forall x, y$
- $d(x, y) = 0$ if $x = y$

b) symmetry

- $d(x, y) = d(y, x), \forall x, y$

c) triangular inequality

- $d(x, z) \leq d(x, y) + d(y, z)$

Example:

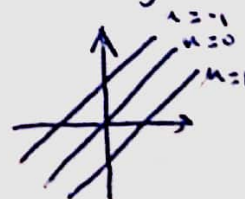
$$f(x, y) = x - y$$

$$L_u = \{(x, y) \in D_f : f(x, y) = u\}$$

$$u = -1 \rightarrow y = x + 1$$

$$u = 0 \rightarrow y = x$$

$$u = 1 \rightarrow y = x - 1$$



Limits

Definition of the limit

$$\lim_{(n,y) \rightarrow (a,b)} f(n,y) = l \Leftrightarrow \forall \epsilon > 0, \exists \delta > 0 :$$

$$0 \leq \|(n,y) - (a,b)\| \leq \delta \Rightarrow |f(n,y) - l| < \epsilon$$

$\underbrace{\quad}_{\text{distance of } (n,y) \text{ to } (a,b)} \quad \underbrace{\quad}_{\text{distance of the function to the limit}}$

Inequalities used at $(0,0)^*$

$$\rightarrow |x|, |y| \leq \sqrt{x^2 + y^2}$$

$$\rightarrow y^2; x^2 \leq x^2 + y^2$$

$$\rightarrow |\sin(n)| \leq 1$$

$$\rightarrow |\cos(n)| \leq 1$$

$$\rightarrow |\sin(n)| \leq |n|$$

* if the limit doesn't go to zero, there should be performed a change of variable in order to use the inequalities above.

Properties of the absolute value:

$$\rightarrow |ab| = |a| \cdot |b|$$

$$\rightarrow \left| \frac{a}{b} \right| = \frac{|a|}{|b|}$$

$$\rightarrow |a+b| \leq |a| + |b|$$

$$\rightarrow |a-b| \leq |a| + |b|$$

$$\hookrightarrow |a+(-b)| \leq$$

$$\leq |a| + |-b| = |a| + |b|$$

Studying limits

To study the existence of a limit at $a \in \mathbb{R}^n$ for a function of several variables f may require a particular study of limits along continuous paths leading to the point a , such as Retas, parabolas (desde que passem pelo ponto desejado, quaisquer Retas serve).

- If f has a limit L at a , then, for any smooth path passing in $x(t)$ $\lim_{t \rightarrow 0} f(x(t)) = L$

- If, for distinct paths x_1 and x_2 beginning in a , we have

$\lim_{t \rightarrow 0} f(x_1(t)) \neq \lim_{t \rightarrow 0} f(x_2(t))$, or that it doesn't exist/

f has no result for a specific path then f has no limit at a .

Continuity

f is continuous at some point $a \in D_f / (a, b) \in D_f$ if

$$\lim_{x \rightarrow a} f(x) = f(a) \quad / \quad \lim_{(x, y) \rightarrow (a, b)} f(x, y) = f(a, b)$$

If a is a limit point of the domain $D \subset \mathbb{R}^n$ of f and $a \in D_f$, if $L = \lim_{x \rightarrow a} f(x)$, we then we might

define the continuous extension of f to $D \cup \{a\}$ as

$$\bar{f}(x) = \begin{cases} f(x) & \text{if } x \in D \\ L & \text{if } x = a \end{cases}$$

Weierstrass Theorem

Given $u \subset \mathbb{R}^n$, we say that u is compact if and only if u is bounded and closed.

If we consider $u \subset \mathbb{R}^n$ such that is compact and $f: u \rightarrow \mathbb{R}$ be a continuous function. Then f attains a maximum and a minimum in u . That is: there exists $x_0, x_1 \in u$ such that $f(x_0) \leq f(x) \leq f(x_1), \forall x \in u$.

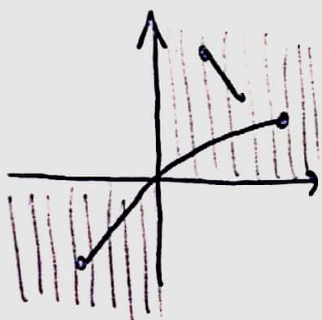
Through the Weierstrass Theorem if we have:

- S , a set compact in $\mathbb{R}^n$
 - f , a continuous function, also, therefore, continuous in S .
- we can conclude that f will have M and m in S .

Boltano Theorem

Given $A \subset \mathbb{R}^n$, we say that A is arc-connected if and only if given $x, y \in A$ there exists a continuous path

$\gamma: [a, b] \rightarrow \mathbb{R}^n$ such that $\gamma(a) = x$, $\gamma(b) = y$, $\gamma(t) \in A$ which is if we can connect all points between the set, then that set is arc-connected, even if the connection of some can only be done by 1 specific path



→ ARC-connected set

Let $A \subset \mathbb{R}^n$ be an arc-connected set and

let $f: A \rightarrow \mathbb{R}$ be a continuous function.

Let $x, y \in A$ be such that $f(x) < f(y)$. Then

for any $u \in]f(x), f(y)[$, there exists $z \in A$ such that $f(z) = u$.

Then through the Boltano's Theorem if we have:

- S , an arc-connected set contained in $\mathbb{R}^n$

- f , a continuous function in $\mathbb{R}$

We can conclude that f will assume all possible values from m to M (considering that the set S ranges from m to M), as there will always be a z in m to M that will have the image we are looking for.

Differential Calculus

Partial Derivatives

$$f: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$(a, b) \in \text{int}(D)$$

$$f_x(a, b) = \frac{\partial f}{\partial x}(a, b) = \lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h}$$

Partial derivative of f in order to x at (a, b)

$$f: D \subset \mathbb{R}^n \rightarrow \mathbb{R}$$

$$a \in \text{int}(D), a = (a_1, \dots, a_n)$$

$$\frac{\partial f}{\partial x_n}(a) = \lim_{h \rightarrow 0} \frac{f(a + h e_n) - f(a)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(a_1, \dots, a_n + h, \dots, a_n) - f(a_1, \dots, a_n)}{h}$$

Partial derivative of f in order to y at (a, b)

$$f_y(a, b) = \frac{\partial f}{\partial y}(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b+h) - f(a, b)}{h}$$

Gradient vector

$$f: D \subset \mathbb{R}^2 \rightarrow \mathbb{R} \quad (a, b) \in \text{int}(D)$$

The gradient vector of f at (a, b) is

$$\nabla f(a, b) = \left(\frac{\partial f}{\partial x}(a, b), \frac{\partial f}{\partial y}(a, b); \underbrace{f(a, b)}_{\text{constant}} \right)$$

The gradient vector is perpendicular to the level curve that passes through (a, b) .

Tangent plane

The equation of the tangent plane that passes through $(a, b, f(a, b))$ is

$$z = f(a, b) + \frac{\partial f}{\partial x}(a, b)(x-a) + \frac{\partial f}{\partial y}(a, b)(y-b),$$

f is an differentiable function on its domain, which (a, b) is an interior point of it.

* If f is a differentiable in $\text{int } D$, we can approximate given value (a, b) to a certain value (approx 1,07; 3,98 to 1; 4) by the tangent plane; applying to it the close point (a, b) the tangent plane at (a, b)

Differentiability

f is differentiable at (a, b) if exist $\frac{\partial f}{\partial x}(a, b)$ and $\frac{\partial f}{\partial y}(a, b)$ and if the function can be approximated by the tangent plane to graph of f at (a, b) that is:

$$\underbrace{f(x, y)}_{\text{value of the function}} = \underbrace{f(a, b) + \frac{\partial f}{\partial x}(a, b)(x-a) + \frac{\partial f}{\partial y}(a, b)(y-b)}_{\text{Value at the tangent plane}} + \underbrace{O(\|(x-a, y-b)\|)}_{\text{error of approximation}}$$

where $\lim_{(x, y) \rightarrow (a, b)} \frac{O(\|(x-a, y-b)\|)}{\|(x-a, y-b)\|} = 0$

So f is differentiable if

$$\lim_{(x, y) \rightarrow (a, b)} \frac{f(x, y) - f(a, b) - \frac{\partial f}{\partial x}(a, b)(x-a) - \frac{\partial f}{\partial y}(a, b)(y-b)}{\|(x-a, y-b)\|} = 0$$

Theorem of Differentiability

If the partial derivatives of a function of f are continuous functions then the function f is a differentiable function. This is that, if by computing the partial derivatives, these result from simple operations between continuous functions then f is differentiable on its domain.

Directional Derivatives

$f: D_f \subset \mathbb{R}^2 \rightarrow \mathbb{R}$, $(a, b) \in \text{int}(D_f)$ and (u, v) a unit vector (norm 1) we have that the directional derivative of f is

$$f'(a, b)_{(u, v)} = \lim_{t \rightarrow 0} \frac{f(a, b) + t(u, v) - f(a, b)}{t}$$

If f is differentiable at (a, b) and (u, v) a unit

vector then: $f'(a, b)_{(u, v)} = \lim_{t \rightarrow 0} \nabla f(a, b) \cdot (u, v) =$

$$= \frac{\partial f}{\partial x}(a,b) \cdot u + \frac{\partial f}{\partial y}(a,b) \cdot v$$

Particular cases
of directional
derivatives

The partial derivatives are particular cases of directional derivatives

$$\frac{\partial f}{\partial x}(a,b) = f'(a,b)_{(1,0)} \quad \text{and} \quad \frac{\partial f}{\partial y}(a,b) = f'(a,b)_{(0,1)}$$

Maximum
value

If f is differentiable at a and $\nabla f(a) \neq \vec{0}$ then the maximum value of the directional derivative at this point is $\|\nabla f(a)\|$ and it occurs in the direction of the vector, $\nabla f(a)$



Maximum value at (a,b)

$$\nabla f(a) = \nabla f(x,y) = (u,v)$$

$$f'(a,b) = \|\nabla f(a,b)\|$$

$$\text{Max} = \|(u,v)\| = \sqrt{u^2 + v^2}$$

$$\frac{\nabla f(a,b)}{\|\nabla f(a,b)\|}$$



$$\|\nabla f(a,b)\|$$

If and only if f is continuous and differentiable at (a,b)



Vector perpendicular to (a,b) with norm = 1



Revisions of anti-derivatives

$$F(x) = \int_a^x f(t) dt$$

$$F'(x) = \left(\int_a^x f(t) dt \right)' = f(x)$$

$$F'(x) = \left(\int_a^{g(x)} f(t) dt \right)' =$$

$$= f(g(x)) \times g'(x)$$

$$\int_a^b f = - \int_b^a f$$

$$\int_a^b f = \int_a^c f + \int_c^b f$$

If it is along another vector (a Random one)

the value of $f'(a,b)_{(u,v)}$

Represents the rate of change of f when we move from (a,b) along the direction of (u,v)

Hessian matrix

↳ matrix of the second order derivatives at $(0,1)$

$$H(x,y) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial f}{\partial x} \left(\frac{\partial f}{\partial y} \right)$$

By the theorem of Schwarz-Young, we can say that

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

Jacobian matrix

considering a function $f(x) = (f_1(x), \dots, f_m(x))$

$f_i: A \rightarrow \mathbb{R}$ such that $\mathbb{R}^n \rightarrow \mathbb{R}^m$

The Jacobian matrix of f will be such that:

$$J_a f = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(a) & \dots & \frac{\partial f_1}{\partial x_n}(a) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(a) & \dots & \frac{\partial f_m}{\partial x_n}(a) \end{bmatrix}_{m \times n} \rightarrow \text{Each row corresponds to the gradient of } f_i \text{ at the point } a.$$

Example:

$$f_1(x, y) = x^2 - 2y^2 \quad f_2(x, y) = 2xy$$

$$\nabla f_1 = (2x, -4y) \quad \nabla f_2 = (2y, 2x)$$

$$J_f(a_1, a_2) = \begin{bmatrix} 2a_1 & -4a_2 \\ 2a_2 & 2a_1 \end{bmatrix}$$

Chain Rule

$$J(g \circ f)(a) = J_g(b) \cdot J_f(a)$$

will be the gradient of g_i at the point $f(a)$

So will be the matrix

$$[\nabla g_i(f(u, v))]$$

Each column will represent the partial derivative of the vector $\frac{\partial f}{\partial x_j}(a)$

$$\begin{bmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} \end{bmatrix} \rightarrow \begin{matrix} \nabla f_1 \\ \nabla f_2 \end{matrix}$$

In order of u In order of v

From here we take that

$$\nabla g_i(f(a)) \cdot \frac{\partial f}{\partial x_j}(a) = \left(\sum_{k=1}^m \frac{\partial g_i}{\partial u_k}(f(a)) \right) \frac{\partial u_k}{\partial x_j}(a) =$$

When it is the form represented on the side

$$\left(= \frac{\partial g}{\partial u} \times \frac{\partial u}{\partial x} + \frac{\partial g}{\partial v} \times \frac{\partial v}{\partial y} = \right)$$

$g < \begin{matrix} u < x \rightarrow \frac{\partial g}{\partial u} \times \frac{\partial u}{\partial x} \\ v < y \rightarrow \frac{\partial g}{\partial v} \times \frac{\partial v}{\partial y} \end{matrix}$

$$= \frac{\partial g}{\partial x_1}(f(a)) \cdot \frac{\partial x_1}{\partial t}(a) + \dots + \frac{\partial g}{\partial x_n}(f(a)) \cdot \frac{\partial x_n}{\partial t}(a)$$

Homogeneous functions

$f: D_f \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ is a homogeneous function of degree α if $f(tx, ty) = t^\alpha f(x, y)$ for every $(x, y) \in D_f$, $t > 0$ such that $(tx, ty) \in D_f$.

Theorem (Euler's theorem)

Let $f: D_f \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be a differentiable function at $(x, y) \in \text{int}(D_f)$. Then f is homogeneous of degree α if and only if

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = \alpha f(x, y)$$

From f being an homogeneous function we

can assume that $f(tx) = t^\alpha f(x)$

And that, if f is differentiable, then the first order partial derivatives are homogeneous of degree $\alpha - 1$ this is

$$\nabla f(tx) = t^{\alpha-1} \nabla f(x)$$

From the chain rule we also get that

$$\nabla f(tx) \cdot x = \alpha t^{\alpha-1} f(x)$$

So if we have a point $g(2, 2) = ?$ and we know that $g(1, 1) = 2$ and g is homogeneous of degree $\alpha = 2$

then $g(2, 2) = g(\overset{t}{2} \times 1, \overset{t}{2} \times 1) = 2^2 \times g(1, 1) = 4 \times 2 = 8$

Taylor's formula

Let $f: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be a C^{n+1} function in an open ball, B , centered in (a, b) and let $(x, y) \in B$. Then:

$$f(x, y) = f(a, b) + \frac{\partial f}{\partial x}(a, b)(x-a) + \frac{1}{2!} \frac{\partial^2 f}{\partial x^2}(a, b)(x-a)^2 + \frac{1}{3!} \frac{\partial^3 f}{\partial x^3}(a, b)(x-a)^3 + \dots + \frac{1}{n!} \frac{\partial^n f}{\partial x^n}(a, b)(x-a)^n + R_n(x, y)$$

$P(x)$ corresponds to $f(x, y)$ without R_n .

With $\lim_{(x, y) \rightarrow (a, b)} \frac{R_n(x, y)}{\|(x-a, y-b)\|^n} = 0$

In $n=2$ we have:

$$f(x, y) = f(a, b) + \overbrace{\frac{\partial f}{\partial x}(a, b)(x-a) + \frac{\partial f}{\partial y}(a, b)(y-b)}^{\text{tangent plane}} + \frac{1}{2!} \left[\frac{\partial^2 f}{\partial x^2}(a, b)(x-a)^2 + 2 \times \frac{\partial^2 f}{\partial x \partial y}(a, b)(x-a)(y-b) + \frac{\partial^2 f}{\partial y^2}(a, b)(y-b)^2 \right] + R_2(x, y)$$

$$\lim_{(x, y) \rightarrow (a, b)} \frac{f(x, y) - P(x, y)}{\|(x-a, y-b)\|^2} = \frac{R_2(x, y)}{\|(x-a, y-b)\|^2} = 0$$

Critical Points

A quadratic form associated to S is:

- positive definite if all elements of the eigen values are positive
- negative if the even order elements of the eigenvalues are positive and the rest negative
- Indefinite for any other combination.

(a, b) is a critical point if $\nabla f(x, y) = (0, 0)$ is equal to (a, b) .

In order to classify the corresponding

critical point it should be study the eigen-
values of the matrix $\nabla^2 f(a, b)$, Hessian matrix
at (a, b) such that

$$\nabla^2 f(a, b) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2}(a, b) & \frac{\partial^2 f}{\partial x \partial y}(a, b) \\ \frac{\partial^2 f}{\partial y \partial x}(a, b) & \frac{\partial^2 f}{\partial y^2}(a, b) \end{bmatrix}$$

principal minors

$$|H_1| = \frac{\partial^2 f}{\partial x^2}(a, b)$$

$$|H_2| = \frac{\partial^2 f}{\partial x^2} \times \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2$$

If the Hessian matrix at (a, b) is positive definite then
 (a, b) is a minimum, if negative then it is a
maximum, and if indefinite a saddle point